SQL SPRINT PROGRAMME - DATABASE FUNDAMENTALS

DDL, DML, DCL & TCL

SQL Command Categories

A complete guide to SQL's four command categories, plus a detailed comparison of DELETE, DROP, and TRUNCATE.

MODULE
Database Commands

FORMAT
Theory + Practice

What this report covers

01 DDL
Data Definition Language - CREATE, ALTER, DROP, TRUNCATE

02 DML
Data Manipulation Language - INSERT, UPDATE, DELETE, SELECT

03 DCL
Data Control Language - GRANT, REVOKE (permissions)

04 TCL
Transaction Control Language - COMMIT, ROLLBACK, SAVEPOINT

Prepared by: Riya Kaldate

The Four Categories of SQL Commands

SQL commands are grouped into four functional categories based on what they do: defining structure, manipulating data, controlling access, or managing transactions. Understanding which category a command belongs to is essential, since each category behaves differently with respect to saving changes (commit behavior).

DDL — Data Definition Language

Defines and modifies database structure (tables, schemas).

`CREATE, ALTER, DROP, TRUNCATE`

DML — Data Manipulation Language

Manipulates the actual data stored within tables.

`INSERT, UPDATE, DELETE, SELECT`

DCL — Data Control Language

Controls access rights and permissions on database objects.

`GRANT, REVOKE`

TCL — Transaction Control Language

Manages transactions - groups of operations treated as one unit.

`COMMIT, ROLLBACK, SAVEPOINT`

DDL - Data Definition Language

DDL commands define, modify, and remove the structure of database objects such as tables, schemas, and indexes. A key characteristic: DDL commands are auto-committed — changes are saved permanently and immediately, and cannot be rolled back.

Key DDL Commands

- CREATE - builds a new table, database, view, or index
- ALTER - modifies the structure of an existing table (add/drop/modify columns)
- DROP - permanently removes an entire table or database object
- TRUNCATE - removes all rows from a table instantly, keeping its structure

DDL_EXAMPLE.SQL

```
CREATE TABLE Students (
  student_id INT PRIMARY KEY,
  name VARCHAR(50),
  age INT
);

ALTER TABLE Students ADD email VARCHAR(100);
```

EXAM TIP

DDL changes are auto-committed - you cannot use ROLLBACK to undo a CREATE, ALTER, or DROP.

DML - Data Manipulation Language

DML commands work with the actual data stored inside tables — inserting new rows, updating existing values, deleting rows, or retrieving data. Unlike DDL, DML changes are NOT auto-committed by default in most database systems and can be rolled back before a COMMIT.

Key DML Commands

- INSERT - adds new row(s) of data into a table
- UPDATE - modifies existing data in one or more rows
- DELETE - removes specific row(s) based on a WHERE condition
- SELECT - retrieves data from one or more tables (read-only, doesn't change data)

DML_EXAMPLE.SQL

```
INSERT INTO Students (student_id, name, age)
VALUES (101, 'Aarav Shah', 20);

UPDATE Students SET age = 21 WHERE student_id = 101;

DELETE FROM Students WHERE student_id = 101;
```

EXAM TIP

DELETE without a WHERE clause removes ALL rows - always double-check the condition first.

DCL - Data Control Language

DCL commands control who can access and perform operations on database objects. These are typically used by database administrators to manage security and permissions for different users or roles.

Key DCL Commands

- GRANT - gives a user specific privileges (SELECT, INSERT, UPDATE, DELETE, etc.) on an object
- REVOKE - removes previously granted privileges from a user
- Privileges can be granted at different levels: a single column, a table, or an entire database
- WITH GRANT OPTION allows the receiving user to further grant the same privilege to others

DCL_EXAMPLE.SQL

```
GRANT SELECT, INSERT ON Students TO 'riya';

REVOKE INSERT ON Students FROM 'riya';

-- 'riya' can now only SELECT, not INSERT
```

EXAM TIP

GRANT gives permissions, REVOKE takes them away - they are direct opposites of each other.

TCL - Transaction Control Language

TCL commands manage transactions — a sequence of DML operations treated as a single unit of work. They let you save changes permanently, undo them, or set checkpoints partway through a transaction.

Key TCL Commands

- COMMIT - permanently saves all changes made in the current transaction
- ROLLBACK - undoes changes made since the last COMMIT (or since the transaction began)
- SAVEPOINT - creates a named checkpoint within a transaction to roll back to partially
- SET TRANSACTION - configures properties of a transaction (e.g. isolation level)

TCL_EXAMPLE.SQL

```sql
UPDATE Students SET age = 22 WHERE student_id = 101;
SAVEPOINT before_delete;
DELETE FROM Students WHERE student_id = 102;
ROLLBACK TO before_delete;
COMMIT;
-- age update is saved; delete is undone
```

EXAM TIP

Once COMMIT runs, changes are permanent - ROLLBACK can no longer undo them.

DELETE, DROP & TRUNCATE - Comparison

These three commands all remove data or objects, but behave very differently in terms of scope, rollback ability, and effect on table structure. Confusing them is one of the most common SQL exam mistakes.

ASPECT	DELETE	DROP	TRUNCATE
Category	DML	DDL	DDL
Purpose	Removes specific rows	Removes entire table/object	Removes all rows, keeps structure
WHERE clause	Yes - can filter rows	Not applicable	Not applicable (removes all)
Rollback	Yes (before COMMIT)	No - auto-committed	No - auto-committed
Speed	Slower (row-by-row, logged)	Fast (drops structure entirely)	Very fast (minimal logging)
Table structure	Stays intact	Completely removed	Stays intact, only data removed
Identity/Auto-increment	Not reset	Removed with table	Reset to initial value

DELETE, DROP & TRUNCATE - Examples

DELETE Remove only students from the 'Sales' department

DELETE.SQL

```
DELETE FROM Students
WHERE department = 'Sales';

-- Removes matching rows only; can be rolled back
```

DROP Permanently remove the entire Students table

DROP.SQL

```
DROP TABLE Students;

-- Table, its data, and structure are all gone
-- Cannot be rolled back
```

TRUNCATE Empty all rows from Students, keep the table structure

TRUNCATE.SQL

```
TRUNCATE TABLE Students;

-- All rows removed instantly; structure remains
-- Auto-increment counter resets to start value
```

REMEMBER

DELETE = some rows (DML). DROP = whole object (DDL). TRUNCATE = all rows, same table (DDL).

Quick Reference & Key Takeaways

The Four SQL Command Categories

- DDL (CREATE, ALTER, DROP, TRUNCATE) - defines structure, auto-committed, cannot be rolled back
- DML (INSERT, UPDATE, DELETE, SELECT) - manipulates data, can be rolled back before COMMIT
- DCL (GRANT, REVOKE) - controls user permissions and access rights
- TCL (COMMIT, ROLLBACK, SAVEPOINT) - manages transactions as units of work

DELETE vs DROP vs TRUNCATE

- DELETE removes specific rows using WHERE, is DML, and is rollback-able
- DROP removes the entire table/object permanently, is DDL, auto-committed
- TRUNCATE removes all rows instantly but keeps the table structure, is DDL, auto-committed

One-Line Cheat Sheet

```
DDL=Structure | DML=Data | DCL=Access | TCL=Transactions
```

Why this matters

Knowing which category a command belongs to tells you instantly whether it can be undone, whether it needs a WHERE clause, and how it affects table structure - all frequently tested exam points.